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Exploring the relationship between productive structure and the informal economy: evidence from Latin American countries

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Abstract

Purpose – The purpose of this study is to examine the relationship between the productive structure and the size of the informal economy in Latin American countries.

Design/methodology/approach – The study employs econometric techniques for panel data covering the period from 2002 to 2017 and considering 17 Latin American countries. The evidence presented is based on the informal economy data generated by Medina and Schneider (2018) who estimate the size of the informal economy using a structural equation model and the share of manufacturing in total employment as a measure of the size of the manufacturing sector. Also, the study addresses the possible endogeneity bias in the relationship studied and makes the conclusions more robust, thus avoiding spurious correlations that weaken the findings.

Findings – The results indicate that most industrialized Latin American countries are associated with a smaller size of the informal economy.

Practical implications – The findings have important policy implications, as they suggest that Latin American economies need to switch the structure of the economy toward more sophisticated productive structures if they want to reduce the size of the informal economy. Thus, more efforts should be deployed to policies to diversify and upgrade economies.

Originality/value – The study contributes to the literature on the informal economy by connecting the country's productive structure and informality. Specifically, the results show that the productive structure of countries is a plausible explanation for the size of the informal economy.

Keywords Productive structure, Developing countries, Informal economy, GMM, GLS

Paper type Research paper

1. Introduction

Currently, a relevant characteristic of societies is the presence of a set of economic activities that operate outside the legal and regulatory frameworks or outside the modern economy (Loayza, 2016). Although the informal economy is a global phenomenon, it prevails in countries with a lower level of development (Loayza, 2016; La Porta and Shleifer, 2014). For example, estimates coming from Medina and Schneider (2018) show that the informal economy reaches an average size of 41% of GDP in developing countries over the period 1991–2017. In the specific case of Latin America, informal activity reached close to 38% of the GDP in the same period. These numbers indicate that informality is an important issue, reflected in the vast literature that has emerged in recent decades around this topic (Williams and Schneider, 2016; Elgin and Erturk, 2019).

The literature has identified a series of potential factors influencing the size of the informal economy, such as GDP per capita (Elgin and Oztunali, 2014; La Porta and Shleifer, 2008, 2014),

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human capital (Docquier *et al.*, 2017), institutional quality (Elgin and Solis-Garcia, 2015; Dau and Cuervo-Cazurra, 2014; Autio and Fu, 2015; Salinas *et al.*, 2020), labor and tax regulations (Botero *et al.*, 2004; Amaral and Quintin, 2006; Cerda and Saravia, 2013), openness to trade (Goldberg and Pavcnik, 2003; Bosch *et al.*, 2012; Paz, 2014), behavioral norms as tax morale (Torgler and Schneider, 2009) among others. These findings have expanded our knowledge of the determinants of the informal economy; furthermore, they reflect the complex and multifaceted nature of the phenomenon.

However, empirical literature connecting a country's productive structure and informality is scarce. The link between a country's productive structure and informal economy can be understood using the lens of the theoretical models of Lewis (1954) and Harris and Todaro (1970). In his seminal work, Lewis (1954) suggests that a migration from a dominant rural sector, based on agriculture, to a more industrialized urban sector is a fundamental step in any process of economic development. The model predicts that as rural–urban migration increases, people's living standards should also increase. Harris and Todaro (1970) extend Lewis model and argue that rural workers do not transit directly to occupations in the modern but to traditional jobs in the urban sector while finding employment in the modern sector. This traditional urban sector corresponds to what is currently called the informal economy. Hence, both theoretical models predict that workers in the informal sector will be absorbed by the jobs created in the modern economy associated with the manufacturing industry (Gutiérrez-Romero, 2021).

In this study, we contribute to the literature on the informal economy and productive transformation by connecting the country's productive structure and informality. More specifically, we try to find whether there is a negative relationship between the size of the informal economy and more sophisticated productive structures, measured by the fraction of employees active in manufacturing. Therefore, the transformation of the structure of the economy is related to its degree of industrialization (Hirschman, 1958; Rodrik, 2013, 2016). To our knowledge, it is among the first studies to examine empirically the nexus between informality and the productive structure using cross-country panel data. The study is built on the literature that argues about economic development as a process of structural transformation of the productive structure, which means a shift of resources from the primary sector to the manufacturing sector and specialized services (Szirmai, 2013; Hidalgo, 2015). Consistent with this literature, we argue that differences in the productive structures of countries affect the evolution and composition of the formal and informal sectors.

We carry out our study using data from 17 Latin American countries (Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Ecuador, El Salvador, Guatemala, Honduras, México, Nicaragua, Paraguay, Perú, República Dominicana, Uruguay and Venezuela) over the period 2002–2017. During the 1950s and 1960s, industrialization occurred in Latin American countries as part of an effort to reduce their primary sectors' dependency, transforming their productive structures. Although most of the Latin American countries followed a common strategy of productive transformation characterized by the industrialization directed at the domestic market, there were significant disparities in the development of manufacturing activities undertaken by the different countries (Ocampo and Ros, 2011). On one hand, Argentina, Brazil, Chile, Colombia and Mexico had a sharp increase in the share of total output and employment corresponding to manufacturing activities. On the other hand, other countries such as Venezuela, Ecuador, Peru and Central America built up modest manufacturing industries (Ocampo and Ros, 2011). Growth in these countries was heavily based on the export of petroleum, bananas, fishmeal and cotton (Hirschman, 1968). As a result of differences in their industrialization processes, there is an important cross-country variation in the productive structure of the Latin American countries.

Latin American countries have recorded steady and robust economic growth related to the 2000s booms in the price of commodities; however, the size of the informal economy has

remained persistently high (Gasparini and Tornarolli, 2009). We argued that this might be explained by the fact that the growth rate in Latin America was not associated with a shift of the economy out of natural resources and into manufacturing or specialized services. By contrast, this region has also experienced a sharp decline in manufacturing output shares in GDP over the last three decades. This phenomenon is known in the literature as premature de-industrialization (Collier and Page, 2009; Rodrik, 2016). For instance, Rodrik (2016) argues that as manufacturing has shrunk informality has grown in Latin American countries. However, he does not provide empirical evidence on this potential link. We fill this gap by empirically testing the existence of a negative relationship between the level of industrial activity of Latin American countries and the size of their informal economies.

To perform this analysis, we use the informal economy data generated by Medina and Schneider (2018) who estimate the size of the informal economy using a structural equation model. This measure includes all economic activities hidden from official authorities for monetary, regulatory and institutional reasons (Medina and Schneider, 2018). In relation to our measure of manufacturing activity, we used the employment share of manufacturing. Using panel data estimation techniques, the results show a robust and negative relationship between the share of employees active in manufacturing and the size of the informal economy. This relationship holds after controlling for a variety of factors that affect the size of the informal economy, such as GDP per capita, human capital, unemployment, tax burden, business regulation, rule of law and corruption. Finally, these findings are robust after conducting some robustness tests and addressing endogeneity issues.

From our perspective, the study is of great interest because it makes three important contributions to the literature. First, by using a non-European sample, our results allow us to establish policy implications focused on the context of developing countries with primary exporting productive stages. Second, we technically address the possible endogeneity bias in the relationship studied and make our conclusions more robust, thus avoiding spurious correlations that weaken our arguments. Third, we fill a gap in the understudied literature, as studies relating the productive structure of countries to the size of their informal economy are scarce.

The rest of the study is organized as follows. Section 2 describes the relationship between productive structure and the informal economy. Section 3 describes the methodology and data. Section 4 presents the findings of the analyses. Section 5 offers several robustness analyses, and the final section represents the conclusion.

2. Literature review

2.1 Productive transformation and economic development

Traditional theories of economic growth consider the endowment and accumulation of factors as the most relevant variables of economic growth (Solow, 1956). However, these ideas were already challenged at the beginning of the 20th century by Schumpeter (1934), for whom economic development is linked to economic change. Particularly, development is related to the capacity of the countries to incorporate new sectors into the economy (Lin, 2011; Reinert, 2007). Likewise, in the 1960s and 1970s, a group of economists working in development studies also questioned the limitations of conventional theories and highlighted the importance of productive transformation and diversification as a necessary condition to achieve economic development (see Rosenstein-Rodan, 1943; Lewis, 1954; Hirschman, 1958; Rostow, 1959).

For example, Murphy *et al.* (1989) affirm that practically all the countries that experienced rapid growth in their productivity and standard of living in the last 200 years have done so through industrialization. To industrialize, countries should gradually move through a series of stages until reaching more complex productive activities, which implies structural change

and productive diversification (Rostow, 1959; Amsden, 1991). The three predominant stages of economic development are the agricultural, the industrial and the high technology services stage. Then, countries should not only seek to have high investment rates but also pay attention to the composition of that investment and the sectors to which it is directed (Hausmann *et al.*, 2007). In summary, the productive transformation is related to the transition process from economies based on natural resources to more sophisticated sectors related mainly to manufacturing and specialized services (Hirschman, 1958; Hidalgo, 2015).

During the 1980s and 1990s, the relevance of productive transformation for economic growth was undervalued by academic circles and policymakers in Latin America as a consequence of the failure of the Import Substitution Industrialization (ISI) model, leading to the so-called lost decade (Hartmann *et al.*, 2016). Despite this, a new research program emerged through authors such as Hidalgo and Hausmann (2009) who have also re-emphasized the role of productive transformation in economic development, under an approach based on economic complexity. By means of network science, these authors demonstrate that economic development is related mainly to greater diversity, rather than a matter of accumulation. Specifically, greater complexity in the products that a country produces implies different results in terms of the development obtained (Hidalgo, 2015). The main argument is that the type of product that a country exports matters for its subsequent economic performance (Hausmann *et al.*, 2007). However, this new line of research is at odds with the ISI model, which advocated for import substitution rather than export promotion (Hidalgo, 2023). For these new policy approaches, the industrial policy should be developed in the context of open economies engaged in export promotion instead of import substitution (Hidalgo, 2023).

Moreover, authors, such as Lin (2011), in their proposal for a New Structural Economy also associate economic development with productive transformation, considering structural changes in technologies and industries as the underlying mechanism. This new approach emphasizes that the countries' development strategy must start from their latent comparative advantages. That is to say that countries must try to identify and pursue comparative advantages that are consistent with their current endowments and over time modify their production structures, allowing them to move toward industries with greater technological complexity. This approach is in line with the findings of Hidalgo and Hausman (2009).

On the other hand, according to the world production network literature, economic development is an "industrial rise" (Gereffi, 1999). To achieve this industrial rise, countries must be inserted into the world system, specifically into the different global production chains (Coe and Yeung, 2015). Industrial raise implies the initial insertion of countries in these global production networks through simple commodities. As more organizational learning is acquired, companies move toward more complex products throughout the value chain. Therefore, countries can achieve economic development if they manage to strategically locate their production in these production networks and over time move toward capital- and knowledge-intensive sectors (Breznitz, 2007, 2021).

Then, according to previously reviewed theories, the shift to manufacturing or specialized services is an indispensable step for sustained economic growth. For example, North *et al.* (2000) argue that the problem of Latin American countries does not lie in the absence of economic growth but in the region's inability to sustain that growth. Unfortunately, Latin American countries are highly exposed to the raw materials price cycles (Ocampo, 2017). Latin American countries have not been able to generate sustained economic growth because they experience stationary economies with little productive transformation, where economic growth has been the result mainly of the adaptive responses of producers to the presence of external shocks, such as a population increase or a change in relative factor prices. We can conclude, then, that economic growth in Latin America is adaptive or exogenous, rather than creative or endogenous from the Schumpeterian perspective (Schumpeter, 1947).

2.2 Productive structure and informal economy

The informal economy is an outstanding characteristic of developing countries, particularly of Latin American countries (Gasparini and Tornarolli, 2009). Over the years, various theories have sought to explain the determinants of the informal economy. However, much of the existing literature on the informal economy builds on dual perspectives of economic development introduced by Lewis (1954). In his seminal work, Lewis suggests that migration from a dominant agriculturally based rural sector to a more industrialized urban sector is a fundamental step in any process of economic development (Busso *et al.*, 2021). As a result, he hypothesized that as rural–urban migration increases, people's living standards should also increase. The basic argument is that agricultural workers would move from an agricultural sector with low productivity and wages to an industrial sector with high productivity and higher wages. In other words, according to Lewis's (1954) model, a shift from the rural sector to the cities would bring about a decrease in the size of the informal economy.

However, Lewis model finds no support in the evidence arising from many developing economies, specifically, in regions such as Latin America. Indeed, between the 1930s and 1970s, Latin American countries attested high rates of rural-to-urban migration (Firebaugh, 1979; Lattes *et al.*, 2004) but rather an increase in the size of the informal economy (Todaro, 1969). As a response to this limitation, Lewis model was expanded by Harris and Todaro (1970) who formulated a two-sectoral model of rural–urban migration, where the urban sector specializes in the production of a manufactured good, while the rural sector specializes in agricultural goods. Unlike the Lewis model, Harris and Todaro (1970) argued that rural workers do not transit directly to employment in the modern sector but to jobs in the traditional urban sector, while finding employment in the modern sector. This implies an excess of supply of workers in the urban sector, many of whom find refuge in the informal sector (Rauch, 1993). Therefore, the Harris-Todaro model suggests the existence of labor segmentation, where there are two urban sectors: formal and informal (Busso *et al.*, 2021).

Inspired by these models, the dual view of informality argues that the informal sector arises as a consequence of the failure of the economy to generate enough productive employment in the formal sector (La Porta and Shleifer, 2008, 2014). For this vision, informal activities are often undertaken out of necessity and in some cases of a temporary nature, due to the lack of employment in the formal sector (Tokman, 1987; Freije, 2002). In addition, informal activities are characterized by the use of primitive technologies and low human capital (Banerjee and Duflo, 2005; La Porta and Shleifer, 2008, 2014; Mondragón-Vélez and Peña, 2010). Based on evidence from developing countries, La Porta and Shleifer (2008) argue that informal firms tend to be smaller and have lower productivity. Looking specifically at Latin American countries, Gasparini and Tornarolli (2009) identify a similar pattern. Therefore, the dual view of informality suggests a segmentation of labor markets, where formal and informal workers coexist, operating in disconnected economic spaces, using different technologies and producing distinct goods (Amaral and Quintin, 2006; La Porta and Shleifer, 2008, 2014; Ulyssea, 2020).

From this dual perspective, faster economic growth is associated with a rapid decline in the size of the informal economy. Therefore, the informal economy will tend to shrink as countries grow (La Porta and Shleifer, 2008, 2014). Nonetheless, many Latin American economies have been growing steadily but have failed to promote a significant mobility from the informal to the formal sector (Rodrik, 2016). A paradigmatic case is Peru, which despite tremendous economic growth over almost two decades, the size of its informal sector remains relatively high (Lust, 2019). In Africa, Chen (2009) shows that 93% of newly created working places, during the 1990s, were part of the informal sector. This hard evidence questions whether it is possible to significantly reduce the size of the informal economy through a mere growth of GDP per capita and opens the possibility to think about alternative mechanisms. For example, the change of economic activities from natural resources to manufactured goods

or even modern services, which are sectors with sufficient capacity to create new jobs, comes up as an alternative pathway. In short, the reduction of the informal sector depends not only on economic growth but also on the quality of growth but rather on what a country produces and exports (Hausmann *et al.*, 2007).

The link between manufacturing and the size of the informal economy can be noticed through various mechanisms. For instance, the manufacturing sector operates with more physical capital and tends to be technologically the most dynamic sector (Knack and Keefer, 1995; Rodrik, 2016). This in turn leads to higher levels of productivity, and therefore, it is possible to provide well-paid jobs (Rodrik, 2011). Indeed, productivity growth is higher in manufacturing than in the rest of the economy (Stollinger *et al.*, 2013; Rodrik, 2016). Additionally, in manufacturing, the optimal plant size of the firms must be sufficiently large to generate increasing returns to scale (Arthur, 1986; Collier and Page, 2009). Consequently, larger firms are less likely to remain under the radar of supervisory authorities (Almeida and Carneiro, 2009; De Paula and Scheinkman, 2011; Ulyssea, 2020). Previous studies show that self-employment is a predominant group in the informal sector (Becker, 2004; Banerjee and Duflo, 2011), while the manufacturing sector is one kind of activity more inclined to formality (Klapper *et al.*, 2010). Moreover, the manufacturing sector in developing countries is usually labor intensive (Collier and Page, 2009), and therefore, it is capable of absorbing labor from the informal sector. Therefore, a productive structure primarily based on natural resources constrains the ability of economies to create good jobs in the formal sector (Blanton and Peksen, 2021), especially in less developed countries (Kpognon, 2022). In fact, Rodrik (2016) argues that the increase in informality in Latin America relates to the decrease in the volume of manufacturing activities. All in all, we hypothesize that differences in the productive structures of Latin American countries lead to variations in the size of the informal economy.

In contrast to what was stated above, in the informality literature the structuralist view points to the existence of important and complex linkages between the formal and informal economy (Moser, 1978; Castells and Portes, 1989) where parts of the production process can be subcontracted or outsourced to other small informal enterprises in the manufacturing sector (Moreno-Monroy *et al.*, 2014; Das, 2017) located in different parts of the world and linked through global production networks (Coe and Yeung, 2015) or also called global value chain (Gereffi *et al.*, 2005). In other words, from the structuralist view, these processes led the informal economy to become a subordinate and dependent sector of the modern capitalist sector (Castells and Portes, 1989). Therefore, increases in outsourcing from the formal sector would be associated with an increase in employment and activities in the informal economy. Our aim in this study is to contrast the validity of these views empirically.

3. Data and methodology

3.1 Data

In this study, we used a cross-section of 17 Latin American countries over the period from 2002 to 2017 ($T = 16$). The selection of countries and years is based on the availability of data. We obtained the data about the informal economy from Medina and Schneider (2018) who used a special type of structural equation modeling to estimate the size of the informal economy (% of GDP) for about 158 countries. This model revises the standard Multiple Indicator-Multiple Cause (MIMIC) approach by using light intensity instead of GDP as an indicator variable (Wu and Schneider, 2021). The database has been widely used in the literature on informality given the robustness of their method and long-period coverage (see Baklouti and Boujelbene, 2020; Colombo *et al.*, 2019; Huynh and Nguyen, 2020; Gutierrez-Romero, 2021). According to this method, the informal economy includes all economic activities that are hidden from official authorities for monetary, regulatory and institutional reasons (Medina and Schneider, 2018). The informal economy is, by nature, difficult to

measure, and all methods, without exception, have advantages and weaknesses (Medina and Schneider, 2019). We acknowledge the drawbacks of the MIMIC model, however, as stated by Medina and Schneider (2021), there is no superior method. Nevertheless, MIMIC estimations lead to robust results comparable with several estimation models (see Medina and Schneider, 2021). Additionally, to check the robustness of the analysis, we use two additional measures on the size of the informal economy. The first alternative measure is an informal economy index constructed by Dau and Cuervo-Cazurra (2014) for 17 countries including our 17 Latin American countries. This index is based on energy consumption. As a second alternative, we use the percentage of the adult population identified as self-employment workers from the International Labour Organization (ILO).

Following the existing literature (Dasgupta and Singh, 2007; Rodrik, 2016; Pierce and Schott, 2016; Haraguchi *et al.*, 2019), we use the share of manufacturing in total employment as a measure of the size of the manufacturing sector. This is often utilized as a crude proxy of a diffuse concept such as industrialization (Gollin *et al.*, 2016). For instance, the degree of deindustrialization is typically understood in the current literature as a decline in the share of manufacturing in a country's total employment (Tregenna, 2009). We obtain these data from the World Development Indicators (WDI) of the World Bank. Moreover, due to the manufacturing sector's capacity to enhance diversification and structural transformations, it is a good proxy for productive transformation (Haraguchi *et al.*, 2019). Similarly, for robustness checks, we use industry gross value added (GVA) as a share of GDP. We also use several control variables which are widely employed in the empirical literature on the informal economy. Thus, we use socioeconomic variables mentioned in the literature such as GDP per cápita (Autio and Fu, 2015; Estrin *et al.*, 2013; Dau and Cuervo-Cazurra, 2014; Salinas *et al.*, 2023), human capital (Tornarolli *et al.*, 2014; Salinas *et al.*, 2020) and unemployment (Thai and Turkina, 2014; Salinas *et al.*, 2019). We obtain these data from the World Bank's WDI and Penn World Tables. The other variables identified in the literature such as tax burden (Webb *et al.*, 2013), rule of law (Dau and Cuervo-Cazurra, 2014; Salinas *et al.*, 2021), business regulation (Klapper *et al.*, 2010; Stenholm *et al.*, 2013) and corruption (Ouédraogo, 2017; Chowdhury, 2005) are related to institutional quality. The data for these measures are obtained from the Heritage Foundation, World Governance Indicators, Fraser Institute and International Country Risk Guide (ICRG), respectively. Table 1 provides variable definitions and sources for the data used in the analysis.

Table 2 shows the descriptive statistics and the correlation matrix of the variables used in this research. First, panel A of this table shows that the data form a balanced panel of 272 observations. Specifically, the panel data have a period of 17 years and a number of 16 Latin American countries. In general, we observe that some variables have greater variability within the time series than when compared across countries. On the other hand, panel B of Table 2 shows the correlation matrix between variables. Here, we can observe that Employment in Industry (EIN), rule of law, business regulation, human capital and our corruption index are negatively correlated with informal activity. Likewise, we observe that the correlations between the independent variables do not exceed 0.60. In other words, we rule out the suspicion of multicollinearity problems among the variables. Table A1 (Appendix) shows the descriptive statistics and correlation matrix for the robustness check variables.

Figure 1 shows quantile maps of the level of Informal Economy (IEC) of Latin American countries and their respective EIN for the first (2002) and last year (2017) of data, respectively. The maps show that informal activity has varied relatively little over time, but there are differences between countries. We observe more significant changes over time for the industry employment variable and also significant differences between countries. In addition, the maps show that the shares of informal activity tend to be higher in countries with lower shares of industry employment. Overall, this suggests a negative pattern between informal activity and industry employment. In other words, industrial activity could be contributing to the formalization of the economy.

Table 1. Description of the variables and data sources

Variable	Symbol	Description	Measure	Data source	Link
Dependent variable					
Informal economy	IEC	Size of the informal economy as a share of GDP normalized between 0 and 1	Proportion	Medina and Schneider (2019) working paper	https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3502028
Independent variable					
Employment in industry	EIN	Percent of a total of people employed in the industrial sector that includes mining, manufacturing, construction and public services (electricity, gas and water)	Percent of total	WDI (modeled ILO estimate)	https://datacatalog.worldbank.org/search/dataset/0037712/World-Development-Indicators
Control variables					
Log (GDP)	Log GDP	GDP per capita, constant 2010 $USD (Log for analysis)	USD per cápita	WDI	https://datacatalog.worldbank.org/search/dataset/0037712/World-Development-Indicators
Tax burden	TB	The measure of the tax burden imposed by a government. It includes direct taxes, in terms of the top marginal tax rates, as a percentage of GDP and normalized between 0 and 1	Proportion	Heritage Foundation	https://www.heritage.org/index/explore?view=by-region-country-year&countryids=&regionids=&yearids=
Rule of law	RL	Perception of the extent to which officers trust the norms of society, property rights, the police and the courts, as well as the likelihood of crime and violence. It is normalized between 0 and 1, where 1 is equivalent to a high rule of law	Index	Worldwide governance indicators	https://www.worldbank.org/en/publication/worldwide-governance-indicators

(*continued*)

Table 1.

Variable	Symbol	Description	Measure	Data source	Link
Business regulation	BR	A measure of the level of administrative requirements and bureaucratic procedures that entrepreneurs must comply with to open or operate a formal business. Normalized between 0 and 1, with 1 equaling the freest business environment	Index	Fraser institute	https://www.fraserinstitute.org/economic-freedom/dataset?geozone=world&year=2021&page=dataset&min-year=2&max-year=0&filter=0&sort-field=regulation&sort-reversed=0
Human capital	HCI	Human capital index based on the average years of schooling	Index	Penn World Tables	https://www.rug.nl/ggdc/productivity/pwt/?lang=en
Corruption	CRT	It is a risk measure that assesses corruption within the political system. This index takes values between 0 and 6, 6 being equal to a high risk of corruption	Index	ICRG	https://borealisdata.ca/dataset.xhtml?persistentId=hdl%3A10864/10120
Unemployment	UN	Percent of total unemployed people	Percent of total	WDI	https://datacatalog.worldbank.org/search/dataset/0037712/World-Development-Indicators
Robustness check variables					
Industry GVA		Measures GVA as a percentage of GDP for mining, manufacturing, construction, electricity, water and gas. Value added is the net output of a sector after adding up all outputs and subtracting intermediate inputs	Percent of GDP	WDI	https://datacatalog.worldbank.org/search/dataset/0037712/World-Development-Indicators
Employment in agriculture		Percentage of total economically active population working in agriculture	Percent of total	WDI	https://datacatalog.worldbank.org/search/dataset/0037712/World-Development-Indicators

(*continued*)

Table 1.

Variable	Symbol	Description	Measure	Data source	Link
Rural population		Rural population refers to the number of people living in rural areas as defined by the national statistical institutes. It is calculated as the difference between the total population and the urban population (log for analysis)	Number of persons	WDI	https://datacatalog.worldbank.org/search/dataset/0037712/World-Development-Indicators
Informal index		It is an index based on the electricity consumption method constructed from the indications of Dau and Cuervo-Cazurra (2014). Specifically, to calculate the informal activity index, we calculate the annual growth rate of electricity consumption and the annual growth rate of GDP. This index represents the growth of informal economic activity from one year to the next and is simply the difference between the total economy index and the formal economy index for a given year. (log for analysis)	Index	WDI	https://datacatalog.worldbank.org/search/dataset/0037712/World-Development-Indicators
Self-employed		Self-employed workers are those workers who, working on their own account or with one or more partners or in a cooperative, perform the type of work defined as "self-employment," i.e., work in which remuneration depends directly on the profits derived from the goods and services produced	Percent of total	WDI (modeled ILO estimate)	https://datacatalog.worldbank.org/search/dataset/0037712/World-Development-Indicators

Source(s): Author's own work

Table 2. Descriptive statistics and correlation matrix of variables

	Informal economy	Employment in industry	Log (GDP)	Tax burden	Rule of law	Business regulation	Human capital	Corruption	Unemp
Panel A: descriptive statistics									
Mean	0.36	20.82	8.52	0.80	0.40	0.05	2.39	2.50	6.79
SD overall	0.11	2.70	0.69	0.06	0.13	0.01	0.33	0.86	3.49
SD between	0.11	2.42	0.70	0.05	0.13	0.01	0.31	0.61	2.97
SD within	0.04	1.33	0.13	0.03	0.03	0.04	0.12	0.62	1.97
Min	0.12	13.34	7.08	0.63	0.09	0.02	1.61	1	2
Max	0.70	27.06	9.61	0.97	0.78	0.07	3.07	5	20.52
N	272	272	272	272	272	272	272	272	272
n	17	17	17	17	17	17	17	17	17
T	16	16	16	16	16	16	16	16	16
Panel B: correlation matrix									
Informality	1								
	–								
Employment in industry	−0.6675*	1							
	(0.0000)	–							
Log (GDP)	−0.7219*	0.3609*	1						
	(0.0000)	(0.0000)	–						
Tax burden	0.3310*	−0.3165*	−0.3875*	1					
	(0.0000)	(0.0000)	(0.0000)	–					
Rule of law	−0.5168*	0.2843*	0.4391*	−0.0589	1				
	(0.0000)	(0.0001)	(0.0000)	(1.000)	–				
Business regulation	−0.3063*	0.0764	−0.0189	0.0678	0.5459*	1			
	(0.0000)	(1.000)	(1.000)	(10.000)	(0.0000)	–			
Human capital	−0.5055*	0.0043	0.5862*	−0.191*	0.5170*	−0.0576	1		
	(0.0000)	(1.000)	(0.0000)	(0.0311)	(0.0000)	(1.000)	–		
Corruption	−0.2350*	0.2791*	0.1502	−0.0905	0.5287*	0.4215*	0.1347	1	
	(0.0012)	(0.0000)	(0.3055)	(1.000)	(0.0000)	(0.0000)	(0.6612)	–	
Unempl	0.3398*	0.0516	0.4434*	−0.306*	0.2775*	0.0418	0.1073	0.1558	1
	(0.0000)	(1.000)	(0.0000)	(0.0000)	(0.0001)	(1.000)	(1.000)	(0.2275)	–

Note(s): *p*-value in parentheses, $^{*}p < 0.05$
Source(s): Author's own work

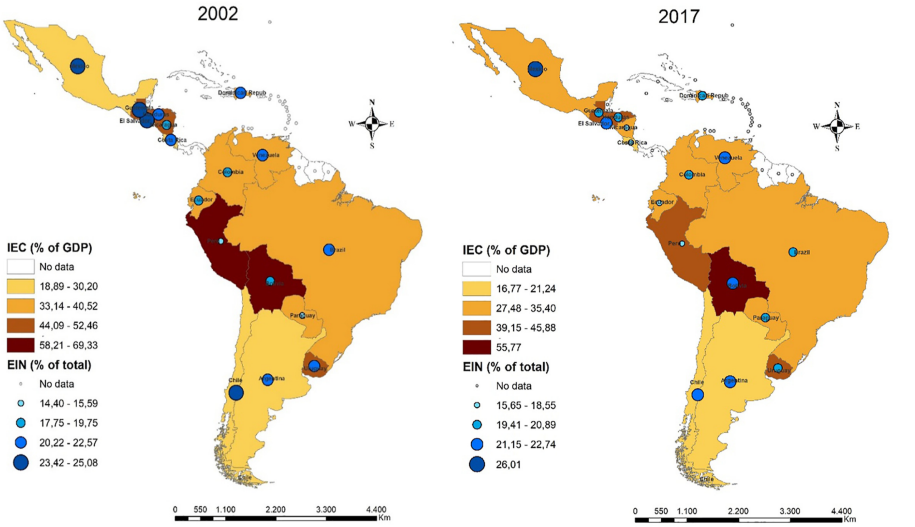


Figure 1.
Informal economy and employment in industry in Latin America

Source(s): Author's own work

3.2 Statistical strategy

In order to analyze the relationship between the size of the informal economy and the productive structure of Latin American countries, we apply panel data regression models. Specifically, we estimate the following model:

$$IEC_{it} = \beta_0 + \beta_1 EIN_{it} + \beta_2 logGDP_{it} + \beta_3 TB + \beta_4 RL_{it} + \beta_5 BR_{it} + \beta_6 HCI_{it} + \beta_7 CRT_{it} + \beta_8 UN_{it} + \varepsilon_{it} + u_i \quad (1)$$

Where $i = 1, 2, \ldots, 17$ refer to countries; $t = 2002, 2003, \ldots, 2017$ refers to years of the data; $\varepsilon_{i,t}$ is the stochastic error term and u_i is the time-invariant individual error term. β_i where $i = 1, 2, \ldots, 8$ are the parameters that measure the correlation between our independent variables and our measure of informal economy.

Before proceeding with estimations, we checked if the use of panel regressions instead of a simple OLS regression is justified. To do so, we applied the Breusch and Pagan test (Breusch and Pagan, 1980). We rejected the null hypothesis (Prob > chibar2 = 0.0000), that is, panel data are preferable to using a pooled dataset. To verify whether it is preferable to use fixed effects (FE) or random effects, we perform a Hausman test (Hausman, 1978). The test consistently suggested that the random effects model is better suited to the data (Prob > chi2 = 0.090). In other words, we reject at the 5% level the non-existence of correlation between the individual effects and the explanatory variables, suggesting the use of the random effects model. We also verified the presence of autocorrelation. We use the test for autocorrelation in panel data derived by Wooldridge (2002). We obtained a result of $F > 0.05$ (Prob > $F = 0.33$), and therefore, the model violates the assumptions of autocorrelation. Also, we applied the Wald test to check for the presence of heteroskedasticity (Juhl and Sosa-Escudero, 2014). The test suggested the presence of the heteroscedasticity problem. To solve these problems, the model will be estimated with random effects using the panel generalized least squares (RE-GLS) estimator with correction for heteroscedasticity and panel-specific AR (1) autocorrelation.

Before starting with the econometric analysis, we checked for serious multicollinearity problems, especially considering that the correlation matrix (Table 2, panel B) shows some correlations above 50% between the independent variables. Table 3 shows the variance inflator factor (VIF) test and we evidence that multicollinearity is not a problem. Specifically, the maximum value of the VIF is 2.79 and the mean value of the VIF test is 1.99, which in any case, shows values below the commonly accepted threshold of 5 and 10 (Cullen *et al.*, 2014; Dau and Cuervo-Cazurra, 2014).

4. Discussion of results

First, in Table 4 we present multiple regression models. In model 1, we run a simple baseline bivariate regression between the share of manufacturing in total employment and the size of the informal economy (informal). Next, we add one control variable at a time to the baseline model 1. In the end, we build a full multivariate model (Model 8). Because it can be argued that the Hausman test (Hausman, 1978) rejects the null hypothesis at a 10% significance level, in Model 9 we show the regression using a FE estimator. We note that when estimating the models using random or FE (Models 8 and 9), the results are consistent in both the signs and magnitude of the coefficients. Models 1–9 include year FE and country FE as control variables. Year FE control for the time effect by including the year variable in the regression, while country FE help control for spatially heterogeneous characteristics across countries. In the case of the FE model (Model 9), country FE are not included, as this estimator removes all time-invariant predictors. Overall, Table 4 shows that the coefficient on our variable of interest, the percentage of manufacturing employment, is negative and statistically significant in all models. Therefore, the negative relationship between employment in the industry and the informal economy is robust to the inclusion of the different control variables in the regression and different estimators. As for the control variables, we find that GDP per capita, rule of law, business regulation, human capital and unemployment are correlated with the size of the informal economy. Moreover, the coefficients of the control variables have the expected signs. The results of the full multivariate model strongly support our theoretical hypothesis that argues for a negative correlation between the share of manufacturing employment and the size of the informal economy. Specifically, the results suggest that most industrialized Latin American countries are associated with a smaller size of the informal economy.

4.1 Robustness checks

Subsequently, as a starting point for the analysis, Model 1 presented in Table 5 is the same model (Model 8) presented in Table 4 for comparative purposes. As mentioned above, industry employment is negatively and significantly correlated with informal activity.

Variable	VIF	SQRT-VIF	Tolerance	*R*-squared
Employment in industry	1.46	1.21	0.6847	0.3153
Log (GDP)	2.79	1.67	0.3581	0.6419
Tax burden	1.48	1.22	0.6769	0.3231
Rule of law	2.34	1.83	0.2991	0.7009
Business regulation	1.80	1.34	0.5543	0.4457
Human capital	2.07	1.44	0.4837	0.5163
Corruption	2.19	1.48	0.4568	0.5432
Unemployment	1.53	1.24	0.6524	0.3476
Mean VIF	*1.99*			

Source(s): Author's own work

Table 3. Test of multicollinearity

Table 4.
Regression analysis

Dependent variable Estimation method	Model 1 Informal RE-GLS	Model 2 Informal RE-GLS	Model 3 Informal RE-GLS	Model 4 Informal RE-GLS	Model 5 Informal RE-GLS	Model 6 Informal RE-GLS	Model 7 Informal RE-GLS	Model 8 Informal RE-GLS	Model 9 Informal FE
Employment in industry	−0.008***	−0.007***	−0.0062***	−0.0062***	−0.006*	−0.006**	−0.005***	−0.005***	−0.0049***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.001)
Log(GDP)		−0.156***	−0.153***	−0.117***	−0.110***	−0.064***	−0.064***	−0.067***	−0.100***
		(0.012)	(0.012)	(0.011)	(0.010)	(0.011)	(0.011)	(0.011)	(0.018)
Tax burden			0.032	0.023	0.010	0.003	0.004	0.006	0.021
			(0.031)	(0.034)	(0.039)	(0.041)	(0.041)	(0.042)	(0.032)
Rule of law				−0.183***	−0.180***	−0.147***	−0.147***	−0.154***	−0.038*
				(0.032)	(0.035)	(0.033)	(0.034)	(0.034)	(0.015)
Business regulation					−0.415	−0.801**	−0.805**	−0.916**	−0.281*
					(0.257)	(0.268)	(0.272)	(0.280)	(0.115)
Human capital						−0.109***	−0.108***	−0.102***	−0.098***
						(0.020)	(0.020)	(0.020)	(0.023)
Corruption							0.000	0.001	0.003
							(0.002)	(0.002)	(0.003)
Unemployment								0.002*	0.001
								(0.001)	(0.001)
Constant	0.498***	1.755***	1.708***	1.468***	1.436***	1.335***	1.335***	1.319***	1.005***
	(0.027)	(0.102)	(0.104)	(0.091)	(0.089)	(0.082)	(0.082)	(0.081)	(0.001)
Observations	272	272	272	272	272	272	272	272	272
N group	17	17	17	17	17	17	17	17	17
chi2	39.410	229.163	229.311	324.359	325.021	493.348	478.204	467.304	–
Autocorrelation tests *p*-value	0.33	0.24	0.18	0.17	0.23	0.41	0.21	0.39	–
Year fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES
Country fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	NO

Note(s): Standard errors in parentheses, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$
Source(s): Author's own work

	Principal model	Robustness checks				
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Dependent variable	Informal	Informal	Informal	Informal	Informal index	Self-employed
Estimation method	RE-GLS	RE-GLS	RE-GLS	RE-GLS	RE-GLS	RE-GLS
Employment in industry	−0.005***				−0.008**	−0.006***
	(0.002)				(0.001)	(0.001)
Industry GVA		−0.002***				
		(0.000)				
Employment in agriculture			0.002***			
			(0.000)			
Rural population				0.004***		
				(0.001)		
Log (GDP)	−0.067***	−0.059***	−0.073***	−0.132***	−0.251***	−0.098***
	(0.011)	(0.010)	(0.009)	(0.015)	(0.038)	(0.008)
Tax burden	0.006	0.026	0.005	0.029	0.467*	0.036
	(0.042)	(0.040)	(0.043)	(0.041)	(0.218)	(0.026)
Rule of law	−0.154***	−0.183***	−0.112**	−0.170***	−0.431**	−0.062*
	(0.034)	(0.033)	(0.037)	(0.031)	(0.165)	(0.029)
Business regulation	−0.916**	−0.918***	−1.706***	−0.833**	−2.409*	−0.254
	(0.280)	(0.267)	(0.366)	(0.264)	(1.173)	(0.193)
Human capital	−0.102***	−0.102***	−0.054**	−0.109***	−0.126	−0.011
	(0.020)	(0.018)	(0.020)	(0.019)	(0.070)	(0.014)
Corruption	−0.000	−0.001	−0.001	−0.000	−0.019	−0.003*
	(0.002)	(0.002)	(0.004)	(0.002)	(0.012)	(0.002)
Unemployment	0.002*	0.002**	0.003**	0.001	−0.004	−0.002**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.005)	(0.001)
Constant	1.319***	1.241***	1.171***	1.560***	5.979***	1.443***
	(0.081)	(0.079)	(0.079)	(0.094)	(0.378)	(0.067)
Observations	272	269	272	272	221	272
N group	17	17	17	17	17	17
chi2	467.304	505.283	604.778	558.829	200.831	509.781
Autocorrelation tests *p*-value	0.23	0.22	0.74	0.13	0.33	0.61
Year fixed effects	YES	YES	YES	YES	YES	YES
Country fixed effects	YES	YES	YES	YES	YES	YES

Note(s): Standard errors in parentheses, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$
Source(s): Author's own work

Table 5.
Robustness checks

As a verification of the robustness of our results shown in Table 5 (Model 2), we employed one alternative measure for the share of employees in manufacturing. Particularly, we use industry GVA. As shown in Model 2 from Table 5, we find a negative and significant relationship between industry GVA and the size of the informal economy. Therefore, the results are consistent with those presented in Model 8. As a further check, we also employed two alternative proxies for productive structure: employment in agriculture and rural population. As displayed in Models 3 and 4 from Table 5, we find a positive and significant relationship. Therefore, the results are consistent with our main hypothesis which suggests that the reallocation of resources from the primary sector to the manufacturing sector would lead to lower informal activity (Lewis, 1954; Todaro, 1969; Harris and Todaro, 1970; Busso *et al.*, 2021). Our main finding is, therefore, robust against specification changes.

For the informal economy, we use two additional measures. First, we estimate an informal economy index taking as a reference the methodology developed by Dau and

Cuervo-Cazurra (2014). These authors employ the energy consumption approach [1]. This method is based on the notion that official GDP can only be considered a good approximation of the measurement of formal economic activity, whereas electricity consumption captures the total economic activity of a country (Dau and Cuervo-Cazurra, 2014). Therefore, an estimate of informal economic activity can be derived by subtracting the formal one from the total economic activity (Kaufmann and Kaliberda, 1996). As a second alternative, we use the percentage of the adult population identified as self-employment workers from the ILO. There is robust evidence for developing economies showing that self-employment is associated with informality (see De Soto, 1989; De Mel *et al.*, 2010; Henley *et al.*, 2009; Williams and Nadin, 2010; Loayza and Rigolini, 2011; La Porta and Shleifer, 2014). The results obtained with the alternative measures are displayed in Models 5 and 6 (Table 5), respectively. As expected, the results are qualitatively similar to Model 1 (Table 5). Therefore, our results are robust to the use of these different estimates of the informal economy. These results are in line with Lust (2019), Blanton and Peksen (2021), and Rodrik (2016) who argue that countries with primary economies and lower levels of industrialization experience higher levels of informal activities, or failing that, lower levels of development associated with informality (Hausmann *et al.*, 2007; Hidalgo, 2015). A similar conclusion has been found for the case of 44 less developed African countries (Kpognon, 2022).

4.2 Endogeneity issues

Subsequently, an important concern in this research is the endogeneity bias caused by at least two factors. The first factor to consider is that there may be an endogeneity bias due to the fact that our variable of interest is correlated with other unobserved factors that may be contained in the error term and that are difficult for us to control (relevant omitted variable). The second factor to consider is the endogeneity bias caused by simultaneous causality between the variables of interest (Hayashi, 2000). Therefore, to address possible endogeneity bias and avoid spurious correlations, we use dynamic panel techniques. First, we use the Generalized Method of Moments (GMM) proposed by Arellano and Bond (1991). This method performs regression with endogenous variables using their differences (the Difference-GMM estimator) as instruments. Second, because it can be argued that our time sample or period (T) is not exactly large, we use the estimator subsequently proposed by Blundell and Bond (1998). This method uses equations with variables in levels and in differences to instrument the endogenous variables (the system-GMM estimator), which increases the size of the matrix (system of equations) and the number of instruments for each variable. In addition to being more advisable for small time samples, since, by incorporating the instruments in levels, it reduces the loss of information, although it could generate over-identification of instruments. The GMM estimators can be implemented in one and two-step approaches. Nevertheless, this study adopted a two-step approach, considering that two-step estimations are more efficient than one-step ones (Windmeijer, 2005). Moreover, in order to control for dynamic panel bias, the $N > T$ condition for adopting the GMM method is satisfied (Roodman, 2009a, b).

Table 6 provide results for Difference-GMM and System-GMM (Models 2 and Model 3, respectively) estimators. Model 1 presented in Table 6 is the same model (Model 1) presented in Table 5 for comparison purposes. The results of the regressions in Models 2 and 3 consistently show that the coefficient of manufacturing employment share is negative and statistically significant, at the level of 5%. This result supports our main hypothesis. Furthermore, we can observe that if we compare the coefficients estimated under the RE-GLS method (which does not address the endogeneity problem of the variable of interest) and the dynamic panel models, our coefficient is overestimated. Therefore, addressing the proposed relationship under a GMM approach seems to be more efficient, since we manage to better

Dependent variable Estimation method	Model 1 Informal RE-GLS	Model 2 Informal Difference-GMM	Model 3 Informal System-GMM
Employment in industry	−0.005***	−0.002*	−0.0035*
	(0.002)	(0.001)	(0.002)
Log (GDP)	−0.067***	−0.198**	−0.269*
	(0.011)	(0.102)	(0.248)
Tax burden	0.006	−0.262	0.270
	(0.042)	(0.328)	(0.465)
Rule of law	−0.154***	−0.151***	−0.152**
	(0.034)	(0.019)	(0.065)
Business regulation	−0.916**	−0.135	−2.537
	(0.280)	(1.153)	(2.244)
Human capital	−0.102***	−0.083	−0.025
	(0.020)	(0.090)	(0.200)
Corruption	0.000	0.002	0.016
	(0.002)	(0.003)	(0.019)
Unemployment	0.002*	0.000	0.000
	(0.001)	(0.002)	(0.007)
Lag.informality		0.496***	0.580**
		(0.087)	(0.231)
Constant	1.319***	1.982*	2.660*
	(0.081)	(1.007)	(1.213)
Observations	272	238	255
N group	17	17	17
chi2	467.304	1096.771	277.194
H0: exogeneity	–	0.01***	0.01***
p-value	–	0.0402	0.0402
Hansen test	–	–	0.193
Sargan test		0.832	–
AR1	–	0.28	0.429
AR2	–	0.64	0.775

Note(s): Standard errors in parentheses, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$
Source(s): Author's own work

Table 6. Difference-GMM dynamic two-step estimation and system-GMM dynamic two-step estimation

approximate a causal effect, which is relatively smaller than the one estimated with the RE-GLS models. This reduction in the magnitude of the coefficient after using GMM models is to be expected, since, by eliminating the correlation between the error term and our independent variable of interest when using instrumental variables, the upward bias of the coefficient is eliminated; thus, the GMM coefficient is reduced (Roodman, 2009a, b). However, although the magnitude of the GMM coefficient is approximately half that of the RE-GLS coefficient, it is still within the 95% confidence interval of −0.00892, −0.00108 estimated in the RE-GLS coefficient.

However, to ensure the validity of the obtained results, we perform several relevant statistical tests. First, we test the null hypothesis of no serial correlation, which is also a test for the validity of instruments. We reject the Arellano and Bond AR (1) at first differences and we do not find evidence for significant second-order autocorrelation (AR (2)). Therefore, the result shows that the model does not suffer from an autocorrelation problem. Furthermore, we employed the Hansen test of over-identifying restrictions that checkthe validity of the instrumental variables. The null hypotheses state that all instruments as a group are exogenous. We fail to reject the null hypothesis, implying that the instrumental variables were valid. Therefore, the results of the Arellano-Bond test and

Sargan's test show that the model is specified correctly and does not suffer from an autocorrelation problem.

5. Conclusions and policy implications

The objective of this study is to examine the relationship between the productive structure and the size of the informal economy in Latin American countries. The Latin American context is considered to be of crucial importance since the evidence points out that Latin American countries have registered steady economic growth related to the commodity price boom in the 2000s; however, the size of the informal economy has remained persistently high (Gasparini and Tornarolli, 2009). To achieve our objective, panel data from the period 2002 to 2017 are used. The analysis covers 17 Latin American countries: Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Dominican Republic, Ecuador, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, Paraguay, Peru, Uruguay and Venezuela. In addition, econometric techniques are used for panel data. In the first instance, sensitivity analysis is performed in order to explore the strength and direction of the relationship between the variables, so that the relationship is robust when we add different control variables widely used in the literature. In the second stage, we use different measures of the countries' production structure to show that the relationship between the variables is robust. In the final stage, with the incentive of solving the endogeneity bias in our explored relationship, and thus avoiding spurious correlations that may weaken our arguments, we use difference-GMM dynamic two-step estimation and system-GMM dynamic two-step estimation.

Our main hypothesis is that there is a negative relationship between the size of the informal economy and more sophisticated productive structures, as measured by the fraction of active employees in manufacturing. Our results confirm the hypothesis by showing that most industrialized Latin American countries are associated with a smaller size of the informal economy. This result can be explained from the point of view of productive transformation, as this transformation is linked to the transition process from natural resource-based economies to more sophisticated and complex sectors, mainly related to manufacturing and specialized services (Hirschman, 1958; Hidalgo, 2015).

The empirical findings of this study offer significant policy implications. For example, our results reveal that Latin American economies need to switch the structure of the economy toward more sophisticated productive structures if reducing the size of the informal economy is wanted. Even when growth rates have been high, Latin American countries have consistently failed to reduce their high rates of informality. A general implication of our results is that the productive structure of countries is a plausible explanation for the achieved size of the informal economy. The import substitution strategy in Latin America, which concentrated efforts on domestic production, showed notable weaknesses in terms of efficiency, competitiveness, and long-term sustainability, contrasting with the case of East Asian countries' success stories such as South Korea and Taiwan in the 1960s and 1970s, where export-oriented industrialization generated sustained growth and rapid structural change (Sheehan, 2010). It suggests that the failure of the Latin American strategy of import substitution to promote productive diversification does not invalidate the arguments defended by the most recent literature based on the structuralist tradition, which continues to stress the importance of industrialization (Lin and Wang, 2020). However, this literature advocates for export promotion rather than import substitution (Rodrik, 2006; Hausmann *et al.*, 2007) and recommends that developing countries follow the dynamic comparative advantages determined by their endowment structure (Lin, 2011.). In accordance with Szirmai (2012), there are no important examples of success in economic development in developing

countries since 1950, which have not been driven by industrialization. Therefore, manufacturing remains critical for development and well-being and has a positive impact on economic growth (Szirmai and Verspagen, 2015; Cantore *et al.*, 2017; Aiginger and Rodrik, 2020).

In seeking to identify policy options to address the informal economy, the findings of our work show that the economic structures of countries matter and that strengthening Latin America's current patterns of comparative advantage, based on natural resources, is not a sustainable long-term strategy. Except for natural resource bonanzas which normally end in severe economic contractions due to plummeting commodity prices (McMillan *et al.*, 2017), there is strong empirical evidence that countries that focus on their manufacturing export sector, particularly for those goods with high technology content, experience higher rates of economic growth than those countries that rely on primary exports (Söderbom and Teal, 2004; Crespo-Cuaresma and Wörz, 2005; Hausmann *et al.*, 2007; Szirmai and Verspagen, 2015). By contrast, economies that rely mostly on natural resources are highly vulnerable to macroeconomic shocks, which are mainly caused by the volatility of international commodity prices (Dasgupta and Singh, 2007; Sheridan, 2014; Ocampo, 2017). Additionally, these findings are consistent with the Resource Curse literature which stresses that natural resource dependency might prevent countries from diversifying their productive structure (Sachs and Warner, 2001; Lederman and Maloney, 2007). Specifically, the Dutch Disease can create incentives for premature deindustrialization (Rowthorn and Ramaswamy, 1997; Rowthorn and Coutts, 2004; Dasgupta and Singh, 2007; Tregenna, 2013; Palma, 2014; Rodrik, 2016). While it is true that natural resources played a major role in the process of economic development in countries like the United States, Australia and Canada in the late 19th and early 20th century, in recent decades studies have found poor growth records of most countries with high resource dependence (Badeeb *et al.*, 2017).

More specifically, Latin American countries must pursue a more agnostic approach to industrial policy aimed at diversifying their productive structures. For instance, the economic complexity approach suggested by Hidalgo and Hausmann (2009) could be a starting strategy. This approach suggests the implementation of development strategies that promote the diversification of products as a way to encourage the accumulation of capabilities (or the creation of new products). These actions, in turn, would encourage a broader productive diversification. Therefore, unlike traditional approaches such as ISI, based on excessive protectionism, our study is in line with Hausmann *et al.* (2007) and Lin (2011), who argue that developing countries should rely more on export-oriented industrialization based on their endowment factors. That is to say that Latin American countries must try to identify and pursue comparative advantages that are consistent with their current endowments and over time modify their production structures, allowing them to move toward industries with greater technological complexity. Moreover, they can achieve economic development if they manage to strategically specialize into specific stages of production following their comparative advantage (Haraguchi *et al.*, 2019) and locate their production in the global value chain, and over time move toward capital- and knowledge-intensive sectors (Breznitz, 2021). Thus, more efforts should be deployed to policies to diversify and upgrade economies.

Finally, we must recognize the limitations and future extensions of this research. For example, this work does not consider a measure of productive diversification or activities located in the service sector. Therefore, future research could extend this study by considering the role of the service sector in economic development. Specifically, the stunning performance of India over the last two decades following an unconventional path of services-led growth as opposed to growth in traditional industry has received considerable interest (Fan *et al.*, 2023). Therefore, it would be worthwhile to investigate whether industrialization is still the most important engine of growth in light of the rise of the service sector. We leave this to future research.

Note

1. For a detailed methodological explanation please see Dau and Cuervo-Cazurra (2014). Additionally, since the World Bank only provides data on electricity consumption from the year 2014, our index is limited to the period from 2002 to 2014. Therefore, Model 6 shows an estimate with 221 observations.

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Appendix

	Informal economy	Industry GVA	Employment in agriculture	Rural population	Informal index	Self-employed
Panel A: descriptive statistics						
Mean	0.36	29.23	19.86	15.15	4.21	0.41
SD overall	0.11	6.41	10.23	1.11	0.47	0.11
SD between	0.11	6.44	10.23	1.14	0.47	0.11
SD within	0.04	2.25	2.39	0.06	0.10	0.02
Min	0.12	18.35	0.05	12.00	2.86	0.22
Max	0.70	53.08	42.39	17.29	5.02	0.67
N	272	269	272	272	221	272
n	17	17	17	17	17	17
T	16	15.82	16	16	13	16
Panel B: correlation matrix						
Informality	1					
	–					
Industry GVA	−0.1463*	1				
	(0.0241)	–				
Employment in agriculture	0.7132*	−0.0293	1			
	(0.0000)	(1.0000)	–			
Rural population	0.0343*	−0.1347	0.1826*	1		
	(0.0000)	(0.4084)	(0.0375)	–		
Informal index	0.4909*	−0.2859*	0.4523*	0.0668*	1	
	(0.0000)	(0.0003)	(0.0000)	(0.0000)	–	
Self-employed	0.7282*	−0.1417*	0.8286*	0.2111*	0.3741*	1
	(0.0000)	(0.0300)	(0.0000)	(0.0069)	(0.0000)	–

Note(s): *p*-value in parentheses, $^{*}p < 0.05$
Source(s): Author's own work

Table A1. Descriptive statistics and correlation matrix for informality and robustness check variables

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